

STUDY OF NETWORK CABLES

EXP No: 02

DATE: 30/7/26

Aim:

Study of different types of network cables.

a) Understand different types of network cable.

Different types of cable used in networking are:

1. Unshielded Twisted Pair (UTP) Cable

2. Shielded Twisted Pair (STP) Cable

3. Coaxial Cable

4. Fibre optic cable

Cable

Type: UTP

Category:

Category 3

Max data Transmission:

100bps

- Category 5

Max data Transmission:

upto 100 Mbps

- Category 5e

Max data Tran: 1 Gbps

Advantages

disadvantages

Advantages

- cheaper in cost
- Easy to install as they have a smaller overall diameter

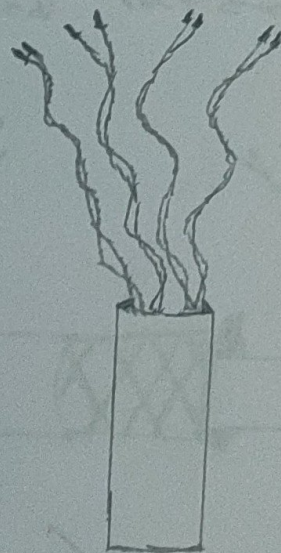
Disadvantages

- More prone to (EMI) Electromagnetic Interference and noise

Application / use

- 10 Base T Ethernet
- Fast Ethernet
- Gigabit Ethernet

Image



2) ~~Cable Type~~ : STP

~~Category~~ :

Category 6, 6a, category 7

Max Data
Transmission :

10 Gbps

10 Gbps

Advantages :

- shielded
- faster than UTP
- less susceptible to noise and interference

Disadvantages :

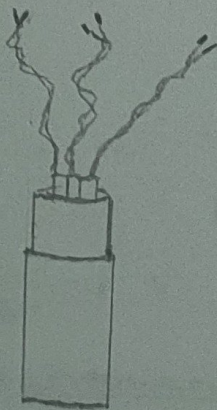
- Expensive

(Greater) installation effort

Application / use

- Gigabit Ethernet-10G Ethernet (55m) widely used in data centres
- Gigabit Ethernet 10G Ethernet (100m)

Image :



3) Cable type : Coaxial cable

category :

RG-6

RG-59

RG-11

Max data Transmission : 10-100 Mbps

Advantages :

- High bandwidth
- Immune to interference
- Low loss bandwidth
- Versatile

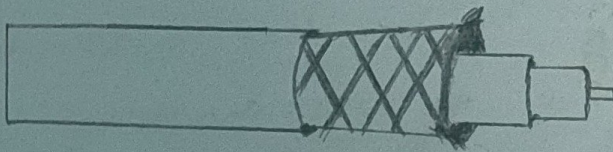
Disadvantages :

- Limited distance
- Cost
- Size is bulky

Application / Use :

- Speed of signal is Television network
- High speed internet connection

Image :



Cable type

Fibre optics cable

Category :

single mode multi mode

Max data

Nonmission 100 Gbps

Advantages :

- High speed
- High bandwidth
- High security
- Long distance

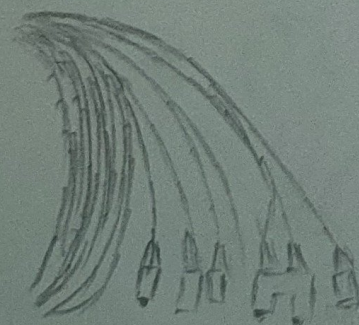
Disadvantages :

- Expensive
- Requires skilled installers

Application :

- Maximum distance of fibre optic cable is around 100 meters

Image :



b) Make Your Own Ethernet Cross Over Cable / Straight Cable

Tools and parts needed :

- Ethernet Cabling. CAT 5e is Certified

Over view
Installation and user interface
To study the packet capture tool

To understand environment of Cisco
Packet to design simple network.

INTRODUCTION :-

A simulator as the name suggests
simulates network device and its
environment.

(1) It allows you to model complex systems
without the need for dedicated
equipment.

(2) It helps you to practice your network
configuration and troubleshooting skills
via computer.

(3) It is available for both Linux &
Windows desktop environments.

Result :-

I have studied the different types
of network cables.